Here we have three mappings of one numerical set onto another. In other words, we have three mappings of an interval onto another interval. In case (a) the interval $[-1,1]$ is mapped onto the interval $[0,1]$; in (b) the interval [[illegible]] is mapped onto the interval $[-1,1]$; and in (c) the interval $[0,\pi]$ is $-\frac{\pi}{2},\frac{\pi}{2}$ mapped onto the interval $[-1,1]$.

What is the difference between mappings (b) and (c), on the one hand, and mapping (a), on the other?

Reader: In cases (b) and (c) we have a one-to-one correspondence, i.e. each point of the set D corresponds to a single point of the set E and vice versa, i.e. each point of E corresponds to only one point of D. In case (a), however, there is no one-to-one correspondence.

Author: Yes, you are right. Assume now that the directions of all the arrows in the figure are reversed. Now, will the mappings define a function in all the three cases?

Reader: Obviously, in case (a) we will not have a function since then the reversal of the directions of the arrows produces a forbidden situation, namely, one number corresponds to two numbers. In cases (b) and (c) no forbidden situation occurs so that in these cases we shall have some new functions.

Author: That is correct. In case (b) we shall arrive at the function $y = \arcsin x$, which is the inverse function with respect to $y = \sin x$ defined over the interval $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. In case (c) we arrive at the function $y = \arccos x$, which is the inverse function with respect to $y = \cos x$ defined over $[0,\pi]$.

I would like to place more emphasis on the fact that in order to obtain an inverse function from an initial function, it is necessary to have a one-to-one correspondence between the elements of the sets D and E. That is why the functions $y=\sin x$ and $y=\cos x$ were defined not on their natural domains but over such intervals where these functions are either increasing or decreasing. In other words, the initial functions in cases (b) and (c) in Fig. 18 were defined as strictly monotonic. A strict monotonicity is a sufficient condition for the above-mentioned one-to-one correspondence between the elements of D and E. No doubt you can prove without my help the following